ICT80004 Weekly Communication – Week #08

Student Name: Arun Ragavendhar Arunachalam Palaniyappan ID: 104837257

Organisation: Commonwealth Scientific and Industrial Research Organisation (CSIRO)

Industry Supervisor: Dr. Shigang Liu

Date Prepared: 26/09/2025 Internship Week #: 8

Day	Date	Task(s) completed
1	Monday 22 Sep 2025 8 hours	○ Implemented a replication pipeline based on the FPA (Familiar Pattern Attack) approach described in the paper provided by our tutor. The pipeline automates generation of small, semantics-preserving code perturbations and validates them against our LLM evaluation harness. ○ Assembled a pilot set of representative functions from the code-vulnerability corpus and produced candidate perturbed versions that preserve runtime behaviour for evaluation. ○ Executed smoke tests of the evaluation harness and prepared batched experiment runs across target models (primary focus: GPT-5). ○ Implemented and tested a set of defensive probes (prompt hardening and simple detection heuristics) to assess mitigation effectiveness.
2	Wednesday 24 Sep 2025 8 hours	○ The literature's core insight — that small, non-functional code perturbations can mislead static-analysis style reasoning in LLMs — is borne out by our smoke tests: perturbed inputs change LLM outputs more often than equivalent clean inputs. ○ Perturbations show transferability tendencies across models in preliminary checks. ○ Basic prompt-hardening reduces some false positives but does not fully eliminate the effect, indicating the need for layered mitigations.

Total hours completed for the week: 16

Plans for next week: #09 week (29 Sept – 03 Oct 2025)

- Run the full pilot: batched experiments across the pilot function set, collect success rates, and produce transferability matrices.
- Correlate FPA susceptibility with code idioms from our earlier vulnerable-code suite to identify high-risk patterns.
- Run comparative mitigation tests (prompt variants + detection heuristics) and document effectiveness.
- Update the Dataset Grid with FPA experiment rows and prepare a short slide summary for the next meeting.

Screenshot of Timely EMAIL communication update to the Supervisor at the end of week #07 sent on 26 September 2025 Friday at 4:38 PM (16:38).

1 of 352

Weekly Communication / Reflection update - #08 week, 22 - 26 Sep 2025

Arun Ragavendhar <arunragavendhar.1999@gmail.com> 16:38 (5 minutes ago)
to Shigang

Dear Dr. Liu,

I hope you are well. Please find my Week 8 update (22–26 Sept 2025) below.

Activities completed this week (Total: 16 hours)

- Implemented a replication pipeline based on the FPA (Familiar Pattern Attack) approach described in the paper provided by our tutor. The pipeline automates generation of small, semantics-preserving code perturbations and validates them against our LLM evaluation harness.
- Assembled a pilot set of representative functions from the code-vulnerability corpus and produced candidate perturbed versions that preserve runtime behaviour for evaluation.
- Executed smoke tests of the evaluation harness and prepared batched experiment runs across target models (primary focus: GPT-5).
- Implemented and tested a set of defensive probes (prompt hardening and simple detection heuristics) to assess mitigation effectiveness.

Key findings / interim observations

- The literature's core insight — that small, non-functional code perturbations can mislead static-analysis style reasoning in LLMs — is borne out by our smoke tests: perturbed inputs change LLM outputs more often than equivalent clean inputs.
- Perturbations show transferability tendencies across models in preliminary checks.
- Basic prompt-hardening reduces some false positives but does not fully eliminate the effect, indicating the need for layered mitigations.

Plan for next week (Week 9: 29 Sept – 3 Oct 2025)

1. Run the full pilot: batched experiments across the pilot function set, collect success rates, and produce transferability matrices.
2. Correlate FPA susceptibility with code idioms from our earlier vulnerable-code suite to identify high-risk patterns.
3. Run comparative mitigation tests (prompt variants + detection heuristics) and document effectiveness.
4. Update the Dataset Grid with FPA experiment rows and prepare a short slide summary for the next meeting.

Kind regards,
Arun Ragavendhar A. P.
ICT80004 Internship Student – CSIRO Data61

Reply Forward